
HW4: PromptIR-Based All-in-One Restoration for Rain and Snow Degraded Images

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GitHub Code Link: <https://github.com/GitHub-WhateverWorks/VIS-HW4>

1 Introduction

Image restoration aims to recover a clean target image from a degraded observation. In this assignment, each degraded image is corrupted by either rain or snow, and the objective is to predict the corresponding clean image. The test images are named only by numeric IDs and do not reveal whether the degradation is rain or snow. Therefore, the model must perform blind restoration: it must infer the degradation characteristics from the input image itself and restore the image without relying on an explicit degradation label.

The assignment requires PromptIR as the restoration model. In this project, PromptIR is treated as a prompt-based restoration framework wrapped around a complete image restoration backbone. The PromptIR component provides degradation-adaptive conditioning through learnable prompts, while the backbone performs feature extraction, multi-scale reconstruction, and residual image prediction. This is suitable for the rain/snow setting because one shared model can restore both degradation types while adapting its internal behavior according to the input image.

For the backbone, I use a Restormer-style multi-scale Transformer encoder-decoder. Restormer is designed for high-resolution image restoration and uses efficient Transformer blocks based on multi-Dconv head transposed attention and gated depthwise feed-forward networks. In my implementation, the network follows a U-Net-like hierarchy: the encoder extracts multi-resolution features, the latent stage captures high-level degradation information, and the decoder reconstructs the clean image using skip connections. PromptIR prompt generation is inserted into this restoration pipeline so that the decoder receives degradation-aware guidance.

All models are trained from scratch using only the provided paired rain/snow dataset. No external dataset, pretrained encoder, pretrained feature extractor, or perceptual network is used. I first trained a completed dim=48 PromptIR-Restormer model as a significant control group. This model reached 27.3 validation PSNR and 29.85 public leaderboard PSNR, and a subsequent 40-epoch full-data fine-tuning stage improved the public score to 30.29. The final main model scales the backbone to dim=64 with crop size 192 and batch size 8 under a two-stage training schedule. This final dim=64 model achieved 30.98 validation PSNR and 31.45 public leaderboard PSNR.

PromptIR is used as the required restoration framework for all-in-one blind image restoration Potlapalli et al. [2023]. The backbone follows the Restormer restoration architecture Zamir et al. [2022]. The training objective uses Charbonnier reconstruction loss Charbonnier et al. [1994].

2 Method

2.1 Dataset and Preprocessing

The dataset contains paired degraded and clean images. The training set has 1600 rain-degraded images and 1600 snow-degraded images, each with a corresponding clean target. The test set contains 100 degraded images named by numeric IDs only, so the model cannot rely on the filename to determine the degradation type.

The preprocessing pipeline is as follows. First, each RGB image is loaded and converted to the $[0, 1]$ floating-point range. During training, paired random crops are extracted from the degraded image and its clean target using the same crop coordinates. This preserves pixel-level correspondence between input and target. The completed dim=48 control mainly uses crop size 160, while the final dim=64 main model uses crop size 192 to provide more spatial context. Random horizontal flipping is used as a safe augmentation because it preserves the natural vertical orientation of rain and snow. During validation and inference, full-resolution images are restored.

A train/validation split is used while searching for architecture and training settings. After selecting a configuration, I also perform full-data fine-tuning using all 3200 training pairs. This removes the validation holdout and gives the model more paired examples during the final polishing stage.

2.2 Overall Training Strategy

The experiments are organized into two groups. The first group is a completed dim=48 PromptIR-Restormer control model. This model is important because it establishes a strong validated baseline and confirms that the PromptIR-Restormer design, prompt length 8, crop size 160, full-data fine-tuning, and x8 test-time augmentation are effective for this dataset. The second group is the final main dim=64 model, which increases the backbone capacity and crop size and replaces the dim=48 control as the final submitted model.

The dim=48 control was trained with base dimension 48, block configuration $[3, 4, 4, 5]$, prompt length 8, crop size 160, and batch size 10. It achieved 27.3 validation PSNR and 29.85 public leaderboard PSNR. A 40-epoch full-data fine-tuning stage further improved the public score to 30.29. This shows that full-data low-learning-rate fine-tuning is useful after the validation-based configuration has been selected.

The final main model uses base dimension 64, prompt length 8, crop size 192, and batch size 8. It is trained with a two-stage schedule. Stage 1 trains from scratch for 180 epochs using learning rate 2×10^{-4} , 15 warm-up epochs, and EMA decay 0.999. Stage 2 fine-tunes for 40 epochs using learning rate 5×10^{-5} , 3 warm-up epochs, and EMA decay 0.9995. Stage 1 reached 30.77 validation PSNR, and Stage 2 further improved the validation PSNR to 30.98. The final public leaderboard score is 31.45.

2.3 Model Architecture

The model follows the PromptIR framework with a Restormer-style backbone. PromptIR is used as a degradation-adaptive prompting mechanism, while Restormer serves as the main restoration backbone. The model is a single all-in-one network shared by both rain and snow images.

Backbone. The backbone is a multi-scale Transformer encoder-decoder. An initial convolution projects the RGB input into a feature tensor with base channel dimension C . The encoder contains multiple resolution levels. After each downsampling operation, the spatial resolution is reduced while the channel dimension increases. The decoder mirrors this process using upsampling and skip connections to recover spatial details. The final output convolution predicts a residual image, which is added to the degraded input to obtain the restored result.

For the dim=48 control model, the channel dimensions are:

$$48 \rightarrow 96 \rightarrow 192 \rightarrow 384.$$

For the final dim=64 main model, the channel dimensions are:

$$64 \rightarrow 128 \rightarrow 256 \rightarrow 512.$$

The block configuration [3, 4, 4, 5] means that the first three encoder levels use 3, 4, and 4 Transformer blocks, while the bottleneck uses 5 Transformer blocks. The decoder mirrors the encoder block allocation.

Transformer Block. Each block follows the Restormer design and consists of two main components: multi-Dconv head transposed attention (MDTA) and a gated depthwise feed-forward network (GDFN). MDTA computes attention across feature channels instead of directly across spatial tokens, which reduces the computational cost for high-resolution restoration. Depthwise convolutions in the query, key, and value projections inject local spatial information. GDFN uses gated feature modulation and depthwise convolution to selectively preserve useful structure while suppressing degradation-related artifacts.

PromptIR Prompt Module. The PromptIR component maintains a set of learnable prompt vectors. In this project, prompt length is set to 8. The prompt generation module computes image-dependent prompt weights from the input feature representation and combines the learnable prompts into a degradation-adaptive prompt. This prompt is then injected into the restoration pipeline to guide the decoder. Since the prompts are generated from image features, the model does not need a rain/snow label at inference time.

Output Head. The output head is a convolutional layer that maps the final decoder feature map back to three RGB channels. The model predicts a residual correction rather than the full clean image directly:

$$\hat{I}_{clean} = I_{degraded} + R(I_{degraded}),$$

where $R(\cdot)$ is the residual predicted by the network.

2.4 Loss Function

The model is trained using a pixel-level reconstruction loss. Since the evaluation metric is PSNR, the loss must encourage accurate pixel reconstruction rather than perceptual realism. I therefore avoid VGG perceptual loss, LPIPS, GAN loss, or any pretrained feature extractor, since these would either hurt PSNR or violate the no-pretrained-weights restriction.

The main reconstruction loss is the Charbonnier loss:

$$\mathcal{L}_{char} = \frac{1}{N} \sum_i \sqrt{(\hat{I}_i - I_i)^2 + \epsilon^2},$$

where $\hat{I}$ is the restored image, I is the clean target, and ϵ is a small constant. Charbonnier loss is a smooth approximation of L1 loss and is commonly used in restoration because it is robust while still preserving pixel accuracy.

2.5 Inference

At inference time, the trained model restores each test image independently and saves the output in the required NPZ format. Each key is the filename, such as `0.png`, and each array is stored in $(3, H, W)$ RGB format as unsigned 8-bit integers.

The final inference setting uses x8 test-time augmentation (TTA). The image is transformed by combinations of horizontal flip, vertical flip, and transpose. Each transformed input is restored by the same model, inverse-transformed back to the original orientation, and averaged. Among the tested inference variants, the last checkpoint with x8 TTA gave the best public leaderboard score, so the same inference strategy is used for the final dim=64 submission.

Table 1: Training configurations used in the completed control model and final model.

Setting	dim=48 control	dim=64 main
Backbone	Restormer	Restormer
Prompt length	8	8
Blocks	[3,4,4,5]	[3,4,4,5]
Crop size	160	192
Batch size	10	8
Stage 1 epochs	180	180
Stage 1 LR	2×10^{-4}	2×10^{-4}
Stage 2 epochs	40	40
Stage 2 LR	5×10^{-5}	5×10^{-5}
TTA	x8	x8
Status	control	final

Table 2: Summary of completed experiments. The dim=48 model is used as a strong control group, while the dim=64 model is the final main model.

Model	Crop	Val PSNR	Public PSNR
dim=48 control	160	27.3	29.85
dim=48 + full-data FT	160	–	30.29
dim=64 Stage 1	192	30.77	–
dim=64 Stage 2 final	192	30.98	31.45

3 Results

3.1 Main Results

The completed dim=48 PromptIR-Restormer model serves as a significant control group. It reached 27.3 validation PSNR and 29.85 public leaderboard PSNR. After the configuration was selected, I performed an additional 40-epoch full-data fine-tuning stage using all available training pairs. This improved the public leaderboard score to 30.29, showing that low-learning-rate full-data fine-tuning is useful after architecture and training settings have been selected.

The final main model is the dim=64 PromptIR-Restormer model. Compared with the dim=48 control, this model increases the backbone channel capacity and uses a larger crop size of 192. Stage 1 reached 30.77 total validation PSNR, with 30.06 PSNR on rain images and 31.54 PSNR on snow images. Stage 2 fine-tuning further improved the total validation PSNR to 30.98, with 30.42 PSNR on rain and 31.60 PSNR on snow. The final public leaderboard score is 31.45, making the dim=64 model the best submitted model.

3.2 Training Curve

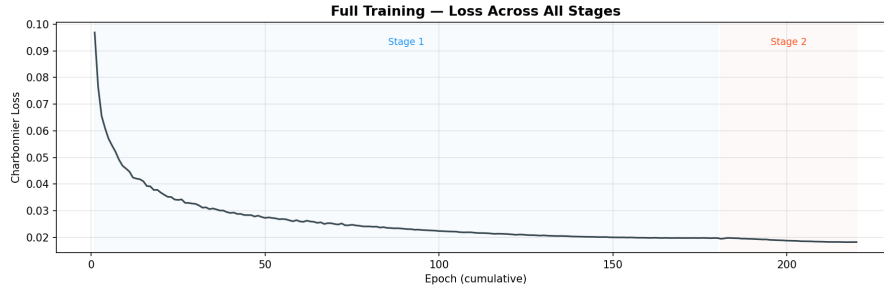
Figure 1 shows the training behavior of the final dim=64 PromptIR-Restormer model. The loss curve measures optimization progress, while the PSNR curve directly reflects restoration quality on the validation split. The two curves show that the model continues to improve after the main training stage, and that the second-stage fine-tuning further refines the restoration output.

3.3 Discussion

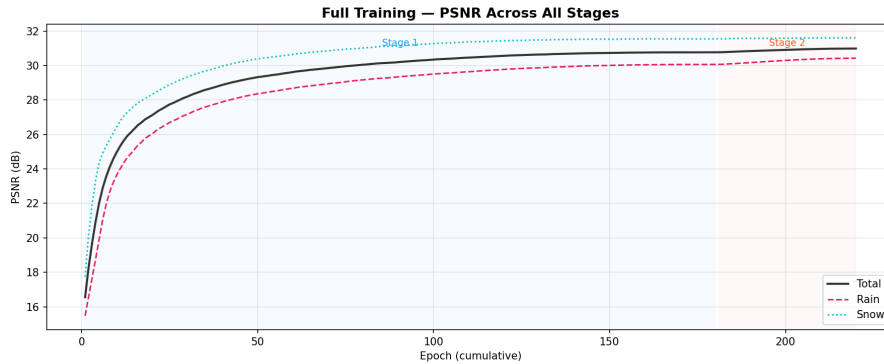
The completed dim=48 control demonstrates that PromptIR with a Restormer backbone is effective for blind rain/snow restoration. However, the final dim=64 model clearly improves over this control group. The improvement supports the hypothesis that increasing backbone capacity is helpful for this task. The larger channel width increases the latent representation

Table 3: Per-degradation validation PSNR of the dim=64 main model.

Stage	Rain PSNR	Snow PSNR	Total PSNR
Stage 1	30.06	31.54	30.77
Stage 2	30.42	31.60	30.98



(a) Training loss curve



(b) Validation PSNR curve

Figure 1: Training curves of the final dim=64 PromptIR-Restormer model. The loss decreases steadily across the main training and fine-tuning stages, while validation PSNR improves from 30.77 dB after Stage 1 to 30.98 dB after Stage 2.

size from 384 channels in the dim=48 model to 512 channels in the dim=64 model, giving the network more capacity to model complex weather degradation patterns.

The per-degradation results show that snow restoration is easier than rain restoration. In Stage 2, the model achieves 31.60 PSNR on snow but 30.42 PSNR on rain. This gap is reasonable because snow particles are usually more localized, while rain streaks are elongated, semi-transparent, and can overlap with image textures. Stage 2 improves both degradation types over Stage 1, but the gain is larger for rain. This suggests that the fine-tuning stage helps the model refine the harder degradation case.

4 Additional Experiments

The assignment states that simply changing hyperparameters does not count as sufficient exploration. Therefore, this section focuses on changes that alter the model behavior, training data usage, or inference mechanism, rather than only changing batch size or learning rate.

Table 4: Effect of prompt length under the dim=48 control setting.

Model	Prompt length	Val PSNR	Public PSNR
dim=48 control	5	26.3	28.70
dim=48 control	8	27.3	29.85

4.1 Increasing Backbone Capacity

Hypothesis. The dim=48 control model already performs well, but rain and snow restoration require modeling both local particles and longer rain streak structures. Increasing the base dimension from 48 to 64 should improve the representation capacity of the Restormer backbone. Because the channel dimensions scale from C to $8C$ in the latent stage, increasing C gives the model substantially more capacity to represent degradation-specific features.

Method. I compared the completed dim=48 control model against a larger dim=64 PromptIR-Restormer model. The dim=48 control uses crop size 160 and batch size 10. The dim=64 main model uses crop size 192 and batch size 8. The larger crop size is used to expose the model to more spatial context, which may help rain removal because rain streaks can span long regions of the image.

Result and implication. The dim=48 control reached 27.3 validation PSNR and 29.85 public PSNR, while full-data fine-tuning improved the public score to 30.29. The final dim=64 model achieved 30.98 validation PSNR after Stage 2 and 31.45 public leaderboard PSNR. This strongly supports the hypothesis that increasing backbone capacity improves restoration quality. The improvement is especially important because it is not only caused by changing optimization hyperparameters; the dim=64 model changes the representational capacity of the PromptIR-Restormer backbone itself.

4.2 Prompt Length Expansion

Hypothesis. PromptIR relies on a prompt dictionary to represent degradation-adaptive information. A smaller prompt dictionary may not be expressive enough to represent both rain streaks and snow particles. Since this task requires one single model to handle two visually different degradations, increasing the prompt length should allow the Prompt Generation Module to learn a richer set of degradation-specific prompt components.

Method. I compared two PromptIR-Restormer models with the same general backbone family but different prompt lengths. The first model used the default prompt length of 5. The second model increased the prompt length to 8. This modification changes the PromptIR prompt module itself: the model has more learnable prompt components available for dynamic degradation conditioning.

Result and implication. The prompt length experiment was conducted under the dim=48 control setting. Keeping the same backbone family and training setup, the prompt length 5 model reached 26.3 validation PSNR and 28.7 public PSNR, while increasing the prompt length to 8 improved performance to 27.3 validation PSNR and 29.85 public PSNR. This corresponds to a gain of approximately 1.0 dB on validation and 1.15 dB on the public leaderboard. The improvement supports the hypothesis that a larger prompt dictionary helps PromptIR represent rain and snow degradations more effectively. This is also why prompt length 8 is used in the final dim=64 model, which achieves 30.98 validation PSNR and 31.45 public leaderboard PSNR.

4.3 Full-Data Fine-Tuning

Hypothesis. After model selection is complete, holding out a validation split reduces the amount of paired training data available to the final model. Fine-tuning on all 3200 training pairs may improve the final leaderboard score because the model receives additional supervised examples.

Method. After selecting the dim=48 control configuration, I fine-tuned the model for 40 additional epochs using the full training set. Validation was disabled during this stage, and the public leaderboard was used only as an external check of the final prediction file.

Result and implication. The full-data fine-tuning stage improved the public score from 29.85 to 30.29. This indicates that additional paired data is valuable once the model configuration has already been selected. This result also supports using a two-stage training procedure for the final dim=64 model, where the second stage performs lower-learning-rate fine-tuning after the main training stage.

4.4 Test-Time Augmentation

Hypothesis. Image restoration predictions can vary slightly under geometric transformations. Averaging predictions from multiple transformed versions of the same input can reduce prediction noise and improve PSNR.

Method. I tested x8 test-time augmentation using horizontal flip, vertical flip, and transpose transformations. Each transformed input is restored by the same model, inverse-transformed back to the original orientation, and averaged.

Result and implication. For the completed dim=48 model, the last checkpoint with x8 TTA produced the best submission among the tested inference variants. Therefore, x8 TTA is also used as the default inference strategy for the final dim=64 model.

5 Conclusion

This project applies PromptIR to all-in-one blind restoration of rain- and snow-degraded images. PromptIR is used as a degradation-adaptive prompting framework, while a Restormer-style Transformer encoder-decoder is selected as the restoration backbone. The completed dim=48 model provides a strong control group, reaching 27.3 validation PSNR and 29.85 public leaderboard PSNR. Full-data fine-tuning further improves this control model to 30.29 public PSNR.

The final main model scales the backbone to dim=64 and uses a larger crop size of 192. This model achieves 30.98 validation PSNR after Stage 2 fine-tuning and obtains a final public leaderboard score of 31.45. The per-degradation results show 30.42 PSNR on rain and 31.60 PSNR on snow, indicating that snow restoration is easier while rain remains the harder degradation type. Overall, the experiments show that PromptIR with a larger Restormer backbone, prompt length 8, full-data fine-tuning, and x8 test-time augmentation is effective for the required single-model rain/snow restoration setting.

References

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